D07 · Warehouse KPIs, Metrics & Governance

01 · Warehouse KPI framework: operational, financial & service metrics

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1 WHAT IT IS

The warehouse KPI framework organizes metrics in three levels — operational (daily), tactical (monthly) and strategic (quarterly) — across receiving, storage, picking, packing, shipping and labor, and links them to cost and service.

2 WHY IT MATTERS

Different audiences need different numbers: supervisors manage lines per hour, managers manage cost per order, executives manage warehouse cost as a share of revenue. **A framework that connects the three levels lets operational improvements show up in financial language (Frazelle — World-Class Warehousing and Material Handling, McGraw-Hill)**, and external benchmarks prevent internal targets from becoming comfortable.

3 HOW IT IS DONE

1. **Choose KPIs per process and level.** A few per process: accuracy, productivity, cycle time, cost.
2. **Document definitions and sources.** One formula and system of record per KPI.
3. **Automate from the WMS.** Dashboards refreshed automatically, not rebuilt in spreadsheets.
4. **Review on a cadence with actions.** Weekly, monthly and quarterly reviews, each ending in owners and next steps.

IN PRACTICE

Report warehouse KPIs at 3 levels — not just the operational level

Supervisors need daily LPPH. The manager needs monthly cost per order. Management needs quarterly warehouse cost % of revenue.

Automate the warehouse dashboard — don’t build it manually

A manually built Excel dashboard has a 3–5 day lag and error risk. A direct WMS connection via Power BI enables automatic updates.

Translate warehouse KPIs into financial language for the executive audience

Directors don’t make decisions based on LPPH — they make decisions based on cost per order and warehouse cost % of revenue.

Compare warehouse KPIs vs. industry benchmark

A 78 LPPH meeting the internal target may be below the sector benchmark (90+ LPPH). Without the external benchmark, internal KPIs can create complacency.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Monthly warehouse dashboard — B2B distribution DC, October management review — All metrics vs. target and vs. previous month.

Warehouse KPI	Current value	Status vs. target
Inventory accuracy % (weekly cycle count)	MEDIUM	97.4% vs. target >99% — degraded, intensified cycle count plan
LPPH (manual + batch picking)	MEDIUM	78 LPPH vs. target >85 LPPH — re-slotting scheduled

Warehouse KPI	Current value	Status vs. target
Perfect order rate %	HIGH	88.4% vs. target 94% — limited by carrier OTIF (91.2%)
Warehouse cost as % of revenue	OK	4.8% vs. target <5% — within target

Result: Warehouse is within cost target but IA is degrading (97.4% vs. 99.2% previous month) requiring immediate attention. Zone A re-slotting scheduled for week 3.

HOW IT IS MEASURED

Warehouse Cost as % of Revenue

Warehouse Cost as % of Revenue(Total warehouse cost / Revenue for the period) × 100

Benchmark: 3–5% in efficient B2B distribution · 5–8% in retail · 6–10% in e-commerce operations with high VAS Warehouse cost % of revenue in an upward trend without channel mix change indicates efficiency loss. Most frequent cause: volume growth without corresponding capacity investment.

Cost per Order (total warehouse cost per order processed)

Cost per OrderTotal warehouse cost for the period / Total orders processed

Benchmark: $4–12/order in efficient B2B distribution · $8–20/order in retail · $15–40/order in e-commerce Cost per order is the most useful financial productivity metric for comparing periods and benchmarks. Declining trend indicates efficiency improvement.

WHAT YOU WOULD USE

Warehouse Analytics

Power BI — Warehouse KPI Dashboard — Module: Automated Warehouse ReportingPower BI dashboard with direct WMS connection: LPPH, IA, pick accuracy, cost per order, and perfect order rate — updated automatically.

SAP EWM Analytics + SAP Analytics Cloud — Module: WMS-Native Reporting + Executive DashboardNative SAP EWM reports for operational KPIs escalating to SAP Analytics Cloud for strategic KPIs.

4 SO WHAT

Measure at three levels and connect them. Few KPIs per process, one definition each, automated dashboards and reviews that always end with an owner and an action.

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